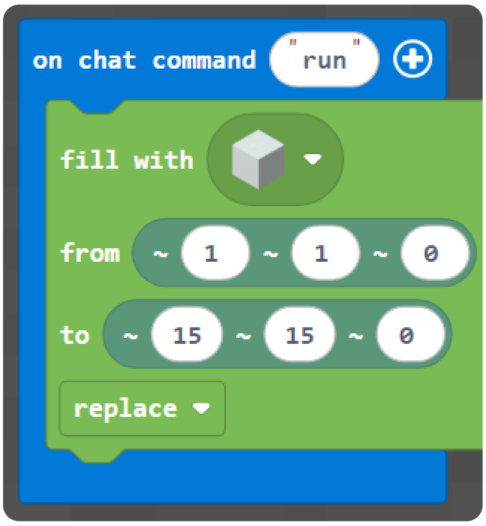
M003 Week 4 — English Worksheet (Advanced)

Topic: Pixel Art 2 — Cat & Tree (x, y) · **Course:** 3D Coordinates · **Level:** Advanced · **Time:** about 45 minutes

This week you copy **two pictures** — a **cat face** and a **tree**. Every block sits at **(x, y)** — x is how far **across**, y is how far **up**. The tricky part is reading shapes that lean and spread, so think before you build.

Build your canvas. Run this to make a blank **15 × 15** wall, then put a **red block at your feet** as your **home spot**:

Blocks



Python

```python
def on_run():
    blocks.fill(WHITE_CONCRETE,
        pos(1, 1, 0),
        pos(15, 15, 0),
        FillOperation.REPLACE)
player.on_chat("run", on_run)
```

> 🔴 **Big idea:** (x, y) is counted from your **home spot**, not the world's corner. Stand on the red block *every* time you run, or the whole picture shifts.

1 · Predict

Read each set of steps. For each one, write what you think you will see **and the reason**.

```
place brown block at (8, 1)
place brown block at (8, 2)
place brown block at (8, 3)
place brown block at (8, 4)
```

What shape is this, across or up? Which number stays the same, and why does that make the line go that way?

```
place orange block at (2, 15)
place orange block at (3, 15)
place orange block at (13, 15)
place orange block at (14, 15)
```

Two pairs of blocks, both at y = 15 but far apart. What part of the cat could each pair be, and why are they in two separate groups?

```
place green block at (8, 15)
place green block at (7, 14)
place green block at (9, 14)
```

These green blocks sit high and spread out as they go down. Do they make a fruit or the top of the tree, and what is happening to x and y at the same time?

```
place green block at (1, 9)
place green block at (1, 10)
place brown block at (8, 1)
place brown block at (8, 2)
```

Two different colours in one run. Describe both lines: which is leaves, which is trunk, and how can you tell from the numbers alone?

2 · Spot the Bug 🐛

Each code block below is broken. Read what it should do, fix it, then explain why the original was wrong and your fix works.

Bug A — This should make the **tree trunk**: four brown blocks going **up** at x = 8, from (8, 1) to (8, 4). Right now they all land in the **same spot**.

```
place brown block at (8, 1)
place brown block at (8, 1)
place brown block at (8, 1)
place brown block at (8, 1)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The tree trunk should grow **up** at x = 8: (8, 2), (8, 3), (8, 4). Right now it grows **across** instead.

```
place brown block at (2, 8)
place brown block at (3, 8)
place brown block at (4, 8)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C (think hard — the numbers are not the problem). A student typed every coordinate correctly, but the whole cat came out **one block too far right and one too low** every time. The code is fine — something about *how* they ran it was wrong.

```
place orange block at (2, 15)
place orange block at (3, 15)
place orange block at (13, 15)
place orange block at (14, 15)
```

What did the student do wrong, and what should they do instead? Explain why the picture shifted even though the numbers were right.

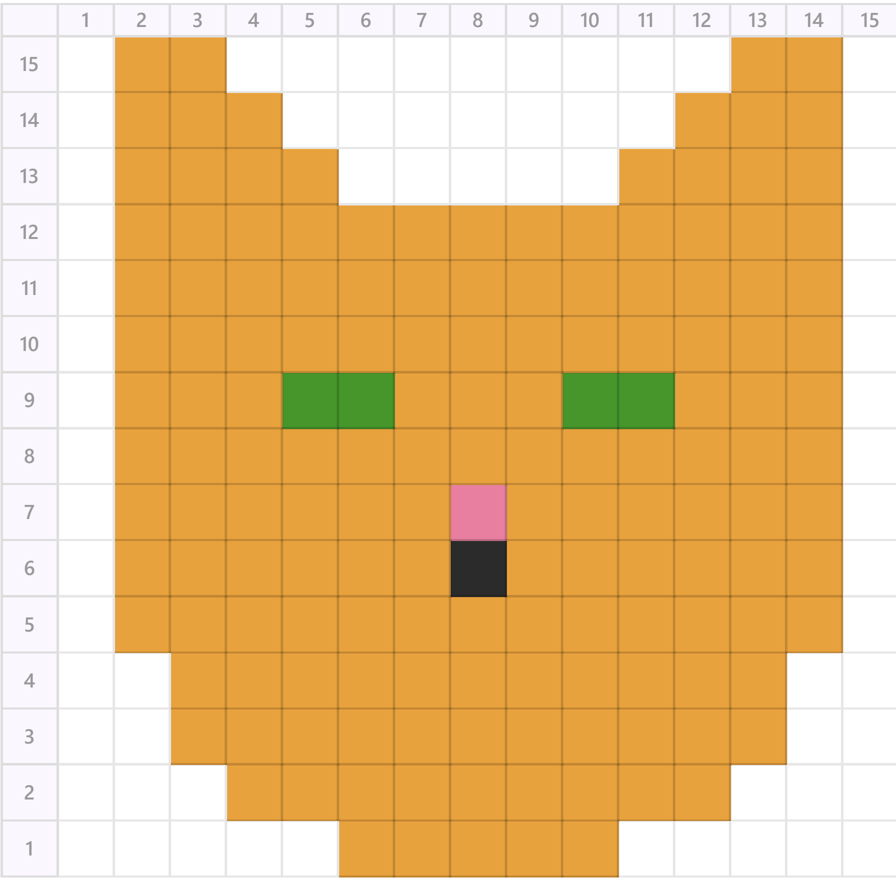
3 · Show Your Work 📸 🎥

Switch to your homework world and stand on your **home spot** (the red block). Plan first, then build. The numbers across the **top** are **x**, the numbers down the **side** are **y**.

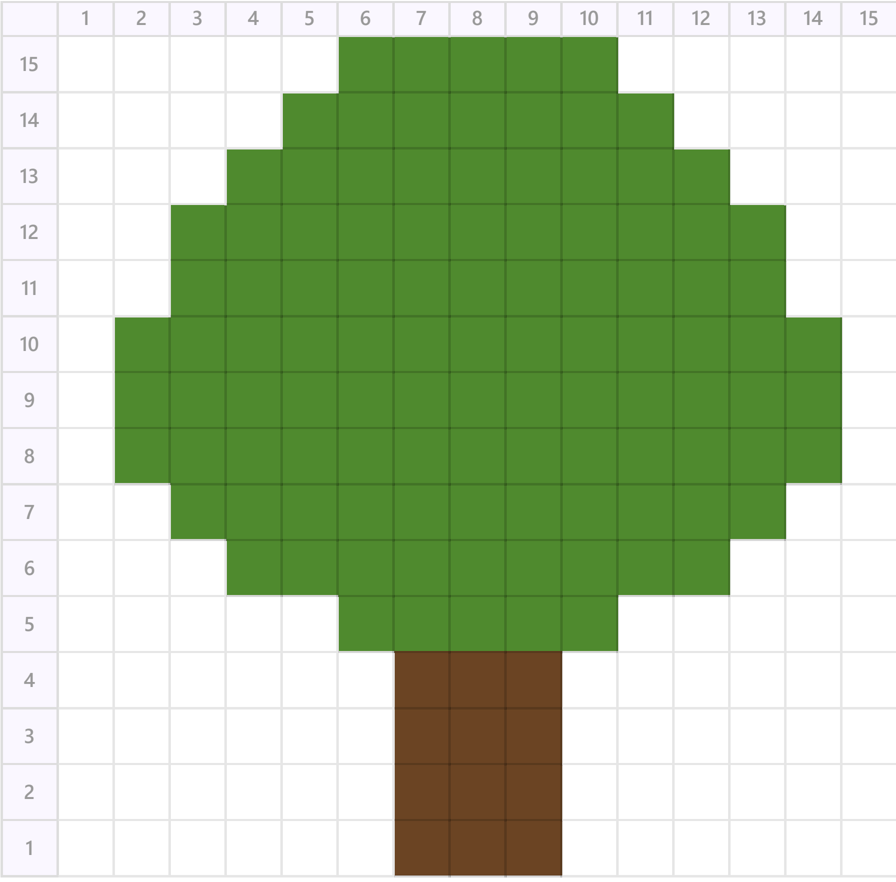
Plan your coordinates first. Before you place anything, write the (x, y) of the **lowest block** and the **highest block** of the picture you will build. This stops the picture running off the top of the wall.

Now build the 😺 cat **and** the 🌳 tree below by reading the (x, y) of every colored square.

😺 Cat face



🌳 Tree



Modify challenge. Once both pictures are built, change one thing with your own idea — give the cat a different eye colour, add an apple to the tree, or make the trunk taller. Build your change, and be ready to say which coordinates you added or swapped.

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

> Today I built _____.
>
> I built it using this code: _____.
>
> In this code I used _____.
>
> The hardest part was _____.
>
> That part was hard because _____.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.